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TRR
JOURNAL OF THE TRANSPORTATION RESEARCH BOARD

Research Article

Transportation Research Record
2023, Vol. 2677(2) 490–504
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Transportation Research Board 2022
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DOI: 10.1177/03611981221107627
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Adopting a Bi-level Optimization Method for the Freight Transportation Problem: A Multi-objective Programming Approach

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Abstract

Nowadays, various companies find it a crucial task to elevate their distribution system efficiency to survive in what is an intensely competitive environment. Freight transportation via road systems may not always be a practical option for the shipper because of financial, social, and environmental aspects. So, in addition to the road system, the shipper can benefit greatly from the rail system. Thus, taking advantage of different transportation modes can help shippers to tackle, to some extent, arising issues. In this paper, we aim at investigating a freight transportation network including rail and road transportation modes in which the shipper's choice affects the decisions deeply. Given that this problem as a bi-level programming model, the rail operator acts as a leader and the shipper is treated as a follower. Once the rail operator reveals the rail transportation price, an appropriate response is received from the shipper and this pattern is repeated until an equilibrium is achieved. From the sustainability view, a multi-objective mixed-integer programming model is developed to simulate their interactions considering a sustainable triple bottom line. Employing the Karush–Kuhn–Tucker approach, we converted the formulated bi-level programming model into a single one. Several numerical examples are investigated to evaluate the validity of the developed model and approaches. Different in-depth sensitivity analyses are carried out and the impacts of several factors, such as line and train capacity, on the rail operator and shipper's decisions are investigated. The results reveal that the rail operator could increase their market share by a wise long-term investment in the rail infrastructure.

Keyword

freight transportation network, mode choice, bi-level programming, stackelberg game, multi-objective optimization

Industrialization and globalization have led to a rapidly great increase in freight transportation in bulk volumes across a vast geographical area. Freight transportation using one transportation mode from origin to destination because of various controversial issues and limitations is not always practical, and that is why concepts such as mode choice and taking advantage of multiple transportation modes in freight transportation have emerged (*1*).

Transportation activities such as cargo transportation, as the second major contributor to carbon emissions (nearly one-fourth of the annual emissions worldwide), have raised concerns across the world. To tackle this important issue, encouraging use of the rail transportation system instead of road transportation is taken into account by authorities. In world-leading economies such as the U.S.A., China, and the European Union (EU), road transportation is a dominant freight mode and has led to an imbalance in the mode split in freight transportation that has eventually given rise to an adverse effects on the environment (*2*, *3*). One of the major drawbacks related to road transportation is noxious vehicle

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emissions of greenhouse gasses (GHGs), which are the main contributor to climate change and the global warming phenomenon. Non-governmental organizations (NGOs) and governments exert intense pressure on factories and transportation companies because of their emissions and, consequently, this pressure has urged companies to find a viable alternative for road transportation to mitigate the adverse effects on the environment. A modal shift from road to rail transportation can not only alleviate this problem, but also it can cut the total costs significantly (*2*).

Previously, road transportation had been the preferred choice but nowadays its capability is limited to some degree because of the disadvantages mentioned earlier (*4*). Emerging concepts, such as multiple transportation modes (e.g., multi-modal and inter-modal transportation), modal shift, and mode choice, could be beneficial to overcome these disadvantages from which road transportation systems suffer (*5*, *6*).

Another applicable transportation mode is employing rail transportation systems. Freight transportation via rail systems is a safe and more economical option compared to road transportation in most cases. Over long-distance transportation for bulk freights, rail systems could be an attractive option. One of the major drawbacks of rail transportation could be the long travel time in comparison to road transportation. To tackle these issues, rail–road transportation provides a practical alternative to road freight transportation (*7*).

Two crucially decisive factors, time and cost, are of great importance for shippers based on their preferences, and two practical transportation options are open to shippers. Based on various aspects, such as price, capacity, travel time, and convenience, they can choose which option should be used and, also, a combination of both modes is available to them (*7*).

In freight transportation, a major issue that may arise is negotiating a price between shippers and carriers, which are in charge of freight transportation. This case can give rise to a game (e.g., a bi-level optimization problem) in which the carrier acts as a leader and can fix a certain price and, subsequently, on the other hand, shippers as followers can respond to the predefined prices.

Stackelberg games are very popular to formulate bi-level optimization problems in many research fields, such as power energy optimization problems (*8–13*), transportation (*2*, *14*, *15*), and so on. Under special conditions, the Karush–Kuhn–Tucker (KKT) method is a capable technique to transform a bi-level optimization problem into a single-level problem effectively (*8*, *10*). In research undertaken by Bahramara et al. (*8*), distribution company (DISCO) is the leader and multiple micro-grids (MGs) are the followers. The DISCO seeks to maximize the total profit and, on the other hand, the MG tries to minimize the total cost. In their work, the KKT approach was employed to transform the bi-level model into a single-level form. A bi-level programming approach for the road pricing and transportation problem was proposed by Lv et al. (*16*), which aims at total emission reduction and congestion mitigation under stochastic and fuzzy environments. To show the applicability of their proposed model and its superiority over traditional approaches, a road pricing problem is employed under numerous policy scenarios. In a soft drink supply chain, a bi-level programming model was developed by Reisi et al. (*17*) in which, in the upper level, two manufacturers and, in the lower level, a retailer, are the main players of a game-based optimization model. A bi-level programming model for a decentralized supply chain including one manufacturer and multiple retailers was developed by Tantiwattanakul and Dumrongsiri (*18*). Considering customers' time-varying demand, the retailer seeks to minimize their costs based on forecasted demand through optimal order and inventory quantity and, on the other hand, the manufacturer aims at minimizing the total cost. For coordination in this supply chain, a non-linear bi-level model is proposed. The non-linear model is converted to a mixed-integer linear programming model and, by employing the KKT approach, the single-level model is obtained.

Another method for dealing with Stackelberg games is using the mathematical program with equilibrium constraints (MPEC). In Wang and Lo (*19*), the authors developed a single-level optimization problem for the road network design problem, which is generally formulated as a bi-level problem or a MPEC. This problem is investigated under the non-convexity conditions arising from the non-linear traffic time function and the traffic assignment equilibrium conditions. In a closed-loop supply chain network design problem under a competitive condition, Yang et al. (*20*) applied a MPEC to simulate a Stackelberg game between entering facilities (leaders) and existing facilities (followers). Integrating a genetic algorithm and an inexact logarithmic-quadratic proximal augmented Lagrangian method, the authors developed a solution method. For solving a Stackelberg game-based network-constrained energy market problem, a mathematical formulation was developed by Gabriel and Leuthold (*21*). In this bi-level problem, a strategic player and an independent system operator act as a leader and follower, respectively. They presumed that there is one strategic player, which leads to a MPEC. Using disjunctive constraints and linearization techniques, this MPEC problem is reformulated as a Mixed-Integer Programming (MIP). Using a MPEC for modeling cyberattacks on the electricity market, Ahmadian et al. (*22*) surveyed a bi-level problem in which an attacker

tries to inject false data into measurement devices, monitored by a system operator. Utilizing the MPEC, the bi-level problem is formulated as a single-level model and, finally, the optimal decision variables of the attacker are achieved.

Investigating travel mode choice is considered a key process in transportation planning to forecast travel demands. In a comparative study (*23*), Hagenauer and Helbich examined mode choice in transportation using seven well-known machine learning algorithms. Mode choice follows an upward trend in most developing countries because of sustainability aspects, such as ever-growing air pollution and congestion in large cities. Giving Taiyuan in China as a case study, Li and Kamargianni (*24*) carried out a research in which the factors affecting mode choice behavior, with particular focus on bike-sharing, are examined carefully. Two key factors affecting the selection of rail transportation are the carbon tax and the low-carbon subsidy. These features, deployed by governments, can put pressure on shippers and encourage rail transport, respectively. In research carried out by Li and Zhang (*2*), in a bi-level programming model, they considered three stakeholders, namely the government, shippers, and rail system operators. At the upper level, the government and the rail operators are considered leaders, while shippers act as followers at the lower level. Because of the complexity of the developed model, they proposed a hybrid algorithm to solve the problem in a timely manner. They demonstrated that the modal shift from road to rail leads to a significant reduction in carbon emissions. An in-depth and statistical analysis was conducted by Regmi and Hanaoka (*25*) to analyze the modal shift from road to rail in the Laem Chabang Port Thanaleng corridor between Laos and Thailand to investigate the reduction of GHG emissions. Their developed mode choice model considering dry port development shows that a modal shift results in a remarkable reduction in emissions.

To the best of our knowledge, limited efforts have been made in employing bi-level optimization in freight transportation, or, more precisely, mode choice in freight transportation problems. We endeavored to formulate a game theory-based interaction between the rail operator and the shipper using a bi-level mixed-integer mathematical programming model. In addition, a sustainable perspective is offered to investigate economic, environmental, and social aspects simultaneously.

The main contributions of this paper can be listed as follows:

- developing a bi-level mixed-integer mathematical programming model to investigate the relationship between the rail operator and the shipper based on the price offered by the rail operator;
- proposing a freight transportation framework consisting of rail and road transportation systems;
- applying a sustainable approach to the discussed problem in this paper, including all three economic, environmental, and social aspects;
- using a multi-objective and the KKT method to transform to single-objective and single-level models, respectively;
- employing several numerical examples to show the model's behavior, and sensitivity analyses on key parameters are conducted;
- utilizing a real case study including data collected from the Iran railway industry and presenting several practical managerial insights.

Mathematical Model

In this section, firstly, the details of the discussed problem are explained, and then mathematical formulations, including notations, objective functions, and constraints, are presented.

Problem Statement

As a freight transportation problem, this problem has two main entities playing key roles. On one hand, the shipper is the owner of transported freight and, on the other hand, the carrier is responsible for transporting freight. In this problem, the shipper faces two options for transporting its goods that are dependent, to a considerable extent, on major factors such as price, transportation time, and capacity. For the shipper, rail and road transportation modes are available: the latter is less time-consuming but the former is to some extent more economical. Based on the shipper's preference, each of the two transportation modes or a combination of them could be chosen. The problem discussed in this paper is considered a bi-level optimization problem in which the railway operator (the carrier) acts as a leader at the upper level and the Ministry of Industry (the shipper) is studied as a follower at the lower level. In other words, a competition is held between two main players in the freight transportation problem. For the sake of simplicity, we occasionally refer to the railway operator and the Ministry of Industry as the leader and follower, respectively. The follower's demand in this problem is demonstrated by origin–destination (OD) pairs, which should be satisfied by transportation modes or be lost.

The leader possesses a heterogeneous fleet of trains with various capacities and initial costs. For the leader, on the one hand, each OD pair has a certain profit, while on the other hand, risk and emission limitations should be taken into account by the leader. For example, one

OD may be considered highly profitable for the leader, but freight transportation on this path may be at risk (e.g., high potential population exposed to hazardous material transportation) or high penalty cost of emissions generated may strongly influence the leader's decisions.

In this problem, the unit transportation price in the railway system is defined by the leader and, subsequently, the follower responds to a predefined price related to the volume of freight transported via the railway system. The leader attempts to maximize total profit as well as minimize total risk and emissions and, on the other hand, the follower aims at minimizing total cost and traveling time. As mentioned earlier, each OD pair has its certain price, risk, and emission and the leader can sort out their priority based on the stated factors.

Assumptions given in the problem explored in this paper are summarized and listed as follows.

- Without loss of generality, it is assumed that lost demand for the customer can occur. In other words, the shipper could be fined for unmet demand by a predefined penalty cost.
- Two transportation modes (railway and road transportation) are considered. To put it another way, other transportation modes (airway and marine) are not included in this problem. The ratio of employing these transportation modes compared to the ones used in this paper is negligible.
- Capacity constraints for trains and blocks are taken into account. The freight transported by each train is limited and trains crossing over certain routes face technical restrictions.
- Constraints on lost demand and rail transportation are imposed in the lower-level model.
- Demand is considered in the form of OD pairs. Each demand is characterized by its origin and destination points and each of the transportation modes or a combination of them can be used for satisfying the demand.
- It is assumed that in this problem, one leader interacts with one follower, in which the former is in charge of rail operation and the latter is responsible for its several subsidiaries.
- In this problem, the shipper acts as a customer desiring to satisfy their demand in an efficient and timely manner. In other words, the emission and risk are not the shipper's concern and the carrier (the upper level) is responsible for the risk and environmental concerns.

The diagram depicted in Figure 1 shows the interconnection between the leader and the follower. As can be seen, the price is determined by the leader at the upper level and is based on the proposed price and other restrictions, and the follower should make an appropriate decision. The made decision by the follower is returned to the leader based on the freight volume transported by the rail system.

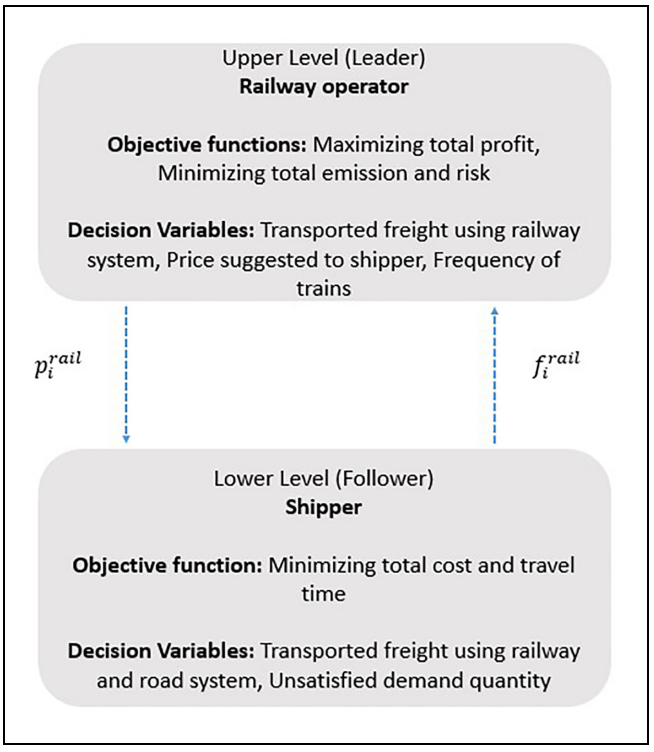


Figure 1. Interaction between leader and follower.

Notation

To aid familiarity with the mathematical formulation for the investigated problem, in this section the sets, parameters, variables, objective functions, and equations are represented.

Sets	
I	Origin–destination (OD) pairs (each OD stands for a certain demand path) $i \in I$
E	Rail tracks (edge) between different stations $e \in H$
T	Available trains for freight transportation $t \in T$
M	Modes of transportation (rail and road) $m \in M$
Parameters	
f^t	Fixed cost for operating train t
Q^t	Volume capacity of train t
vc_i^t	Variable cost of train t for demand path i
D^i	Demand on demand path i
θ^{lost}	Penalty cost for lost demand
p_i^{road}	Price proposed by road transportation for demand path i
ϑ^e	Line capacity for edge e

(continued)

(continued)

Parameters	
em^i	Total emission from freight transportation for demand path *i* by the rail system
r^i	Risk associated with freight transportation for demand path *i* by the rail system
φ^{ti}	Incidence matrix: 1 if train *t* can service demand path *i*, 0 otherwise
Pr^i	Probability of hazardous incidence in demand path *i* by rail
pop^i	Number of people exposed to hazardous incidence in demand path *i* by the rail system
τ^{te}	Incidence matrix: 1 if train *t* passes edge e, 0 otherwise
td_i^m	Travel duration for path *i* using transportation mode *m*
$R_i^{\max}$	Maximum capacity of rail transportation system for demand path *i*
$L_i^{\max}$	Maximum acceptable lost demand by shipper for demand path *i*

Variables	
p_i^{rail}	Price proposed by rail operator for OD *i*
f_i^t	Freight volume transported by train *t* for OD *i*
f_i^{lost}	Quantity of lost demand for OD *i*
f_i^m	Freight volume transported by transportation mode *m* (rail or road) for OD *i*
y^t	Frequency of train *t*

Mathematical Formulation

As denoted earlier, our problem is treated as a bi-level programming model in which the rail operator is the leader (at the upper level) and, at the lower level, the shipper acts as the follower. Firstly, upper-level formulations including objective functions and constraints are presented and, similarly, the follower's formulations are proposed afterward.

Upper Level.

$$\text{Max } Z_1 = \sum_i p_i^{rail} f_i^{rail} - \sum_t f^t y^t - \sum_{i,t} vc_i^t f_i^t \quad (1)$$

$$\text{Min } Z_2 = \sum_i r^i f_i^{rail} = \sum_i Pr^i pop^i f_i^{rail} \quad (2)$$

$$\text{Min } Z_3 = \sum_i em^i f_i^{rail} \quad (3)$$

In Equation 1, the leader's objective function seeks to maximize total profit, comprising the total revenue of freight transportation minus total costs, including fixed train costs and variable freight transportation costs. The second objective in Equation 2 aims at minimizing the total risk associated with each OD pair. Minimizing GHG emissions from rail transportation is controlled by the third objective, function 3 in Equation 3:

$$\sum_t f_i^t \leq D^i \quad \forall i \quad (4)$$

$$\sum_i \varphi^{ti} f_i^t \leq Q^t y^t \quad \forall t \quad (5)$$

$$\sum_t \tau^{te} y^t \leq \vartheta^e \quad \forall e \quad (6)$$

$$0 \leq p_i^{rail} \leq p^{road} \quad \forall i \quad (7)$$

$$\sum_t f_i^t = f_i^{rail} \quad \forall i \quad (8)$$

$$f_i^t \geq 0 \text{ and } y^t \in \text{Integer} \quad \forall i, t \quad (9)$$

Constraint 4 shows that the total freight transported through various trains must be less than the customer's demand. Constraints 5 and 6 present restrictions on trains and rail tracks, respectively. Constraint 7 imposes an upper bound for rail transportation prices. Equation 8 shows the balance between rail freight transportation and trains. Constraint 9 presents the domain of the decision variables.

Lower Level. Similar to the upper level, the lower level has an objective function and several constraints:

$$\text{Min}Z' = \alpha \left[\sum_i p_i^{rail} f_i^{rail} + \sum_i p_i^{road} f_i^{road} + \sum_i \theta^{lost} f_i^{lost} \right] + (1-\alpha) \left[\sum_{i,m} td_i^m f_i^m \right] \quad (10)$$

The follower side seeks to minimize the total cost as well as total transportation time, based on their priorities and preferences according to Equation 10. The follower's total cost is made up of three main terms containing transportation costs related to rail, road, and lost demand, respectively:

$$f_i^{rail} + f_i^{road} + f_i^{lost} = D^i \quad \forall i \quad (11)$$

$$0 \leq f_i^{lost} \leq L_i^{\max} \quad \forall i \quad (12)$$

$$0 \leq f_i^{rail} \leq R_i^{\max} \quad \forall i \quad (13)$$

$$_i^{road} \geq 0 \quad \forall i, m \quad (14)$$

Equation 11 denotes that customer demand can be satisfied through rail or road modes or could be lost to some degree based on decisive factors such as penalty cost, train capacity, and so forth. Constraints 12 and 13 express limitations in unsatisfied demand and rail transportation capacity, respectively. Constraint 14 specifies the domain of the rail transportation decision variables.

Solution Method

After developing a bi-level optimization model for the freight transportation network, this problem must be solved. On account of the bi-level nature of our problem, it cannot be solved directly with solvers. The KKT method has shown its capability in solving such problems. In addition to the KKT method, a fuzzy multi-objective approach is utilized to convert the multi-objective model into a single one. To this end, we use these methods to handle the developed multi-objective bi-level models. In the following sections, the details of employing the KKT method and multi-objective approach are described in detail.

KKT Condition

By using the KKT approach, the follower's problem can be shifted to the upper level because it is linear and convex. For this aim, the follower's constraints are written as follows:

$$C_i^1 = f_i^{road} \geqslant 0 \ \pi_i^1 \ \forall i \tag{15}$$

$$C_i^2 = f_i^{rail} \geqslant 0 \ \pi_i^2 \ \forall i \tag{16}$$

$$C_i^3 = f_i^{lost} \geqslant 0 \ \pi_i^3 \ \forall i \tag{17}$$

$$C_i^4 = L_i^{\max} - f_i^{lost} \geqslant 0 \ \pi_i^4 \ \forall i \tag{18}$$

$$C_i^5 = R_i^{\max} - f_i^{rail} \geqslant 0 \ \pi_i^5 \ \forall i \tag{19}$$

$$f_i^{rail} + f_i^{road} + f_i^{lost} = D^i \ \mu_i \ \forall i \tag{20}$$

$$C_i^j \geqslant 0 \ \forall j = 1, \ \ldots, \ 5 \tag{21}$$

$$\pi_i^j \geqslant 0 \ \forall j = 1, \ \ldots, \ 5 \tag{22}$$

$$C_i^j \times \pi_i^j = 0 \ \forall j = 1, \ \ldots, \ 5, \ i \tag{23}$$

$$\mu_i \text{ unrestricted in sign} \tag{24}$$

Because of multiplying C_i^j and π_i^j, Equation 23 is considered non-linear and can be easily transformed into a linear form by introducing a new binary variable called U_j:

$$C_i^j \leqslant M \times U_j \ \forall j = 1, \ \ldots, \ 5, \ i \tag{25}$$

$$\pi_i^j \leqslant M \times \left(1 - U_j\right) \ \forall j = 1, \ \ldots, \ 5, \ i \tag{26}$$

Now, the Lagrangian function based on the follower's objective function and constraint is constituted as follows:

$$\begin{aligned} L = {} & \alpha \left[\sum_i p_i^{rail} f_i^{rail} + \sum_i p_i^{road} f_i^{road} + \sum_i \theta^{lost} f_i^{lost} \right] + \\ & (1-\alpha) \left[\sum_{i,m} td_i^m f_i^m \right] - \pi_i^1 f_i^{road} - \pi_i^2 f_i^{rail} - \pi_i^3 f_i^{lost} \\ & - \ \pi_i^4 \left(L_i^{\max} - f_i^{lost}\right) - \pi_i^5 (R_i^{\max} - f_i^{rail}) \\ & - \mu_i \ \left(f_i^{rail} + f_i^{road} + f_i^{lost} - D^i\right) \end{aligned} \tag{27}$$

After formulating the Lagrangian function, the following relations are employed for solving the bi-level problem:

$$\frac{\partial L}{\partial f_i^{rail}} = \alpha p_i^{rail} + (1-\alpha) t_i^{rail} - \pi_i^2 + \pi_i^5 - \mu_i = 0 \ \forall i \tag{28}$$

$$\frac{\partial L}{\partial f_i^{road}} = \alpha p_i^{road} + (1-\alpha) t_i^{road} \ - \pi_i^1 - \mu_i = 0 \ \forall i \tag{29}$$

$$\frac{\partial L}{\partial f_i^{lost}} = \alpha \theta^{lost} - \pi_i^3 + \ \pi_i^4 - \mu_i = 0 \ \forall i \tag{30}$$

In the leader's objective function, Equation 1, because of the non-linear term $p_i^{rail} f_i^{rail}$, we should replace this non-linear term with a linear one. For this aim, we utilize the strong duality theory, as follows:

$$\text{Max } D^{Follower} = \ - \sum_i \pi_i^4 L_i^{\max} - \sum_i \pi_i^5 R_i^{\max} + \sum_i \mu_i D^i \tag{31}$$

$$\pi_i^1 + \mu_i \leqslant \alpha p_i^{road} + (1-\alpha) t_i^{road} \ \forall i \tag{32}$$

$$\pi_i^2 - \pi_i^5 + \mu_i \leqslant \alpha p_i^{rail} + (1-\alpha) t_i^{rail} \ \forall i \tag{33}$$

$$\pi_i^3 - \pi_i^4 + \mu_i \leqslant \alpha \theta^{lost} \ \forall i \tag{34}$$

Based on the strong duality theory, the optimal value of an objective function in a primal problem is equal to the corresponding dual objective function value. We implement this theory for the follower's primal and dual problems. Dual formulation of the follower problem can be written as Equations 31–34:

$$D^* = Z'^* \tag{35}$$

$$\begin{aligned} & - \sum_i \pi_i^4 L_i^{\max} - \sum_i \pi_i^5 R_i^{\max} + \sum_i \mu_i D^i \\ & = \alpha \left[\sum_i p_i^{rail} f_i^{rail} + \sum_i p_i^{road} f_i^{road} + \sum_i \theta^{lost} f_i^{lost} \right] \\ & + (1-\alpha) \left[\sum_{i,m} td_i^m f_i^m \right] \end{aligned} \tag{36}$$

$$\frac{1}{\alpha}\left[\left[-\sum_i \pi_i^4 L_i^{\max} - \sum_i \pi_i^5 R_i^{\max} + \sum_i \mu_i D^i\right] - (1-\alpha)\left[\sum_{i,m} td_i^m f_i^m\right]\right] - \left(\sum_i p_i^{road} f_i^{road} + \sum_i \theta^{lost} f_i^{lost}\right) = \sum_i p_i^{rail} f_i^{rail} \tag{37}$$

By using the results obtained from the strong duality theory, the non-linear term of the first objective function of the leader's problem can be linearized as follows:

$$\text{Max } Z_1 = \sum_i p_i^{rail} f_i^{rail} - \sum_t f^t y^t - \sum_{i,t} v_i^t f_i^t \tag{38}$$

$$= \frac{1}{\alpha}\left[\left[-\sum_i \pi_i^4 L_i^{\max} - \sum_i \pi_i^5 R_i^{\max} + \sum_i \mu_i D^i\right] - (1-\alpha)\left[\sum_{i,m} td_i^m f_i^m\right]\right] \tag{39}$$

Figure 2 shows a brief outline of the above-mentioned mathematical formulations, and consecutive steps taken to convert a bi-level mathematical formulation into a single one are shown.

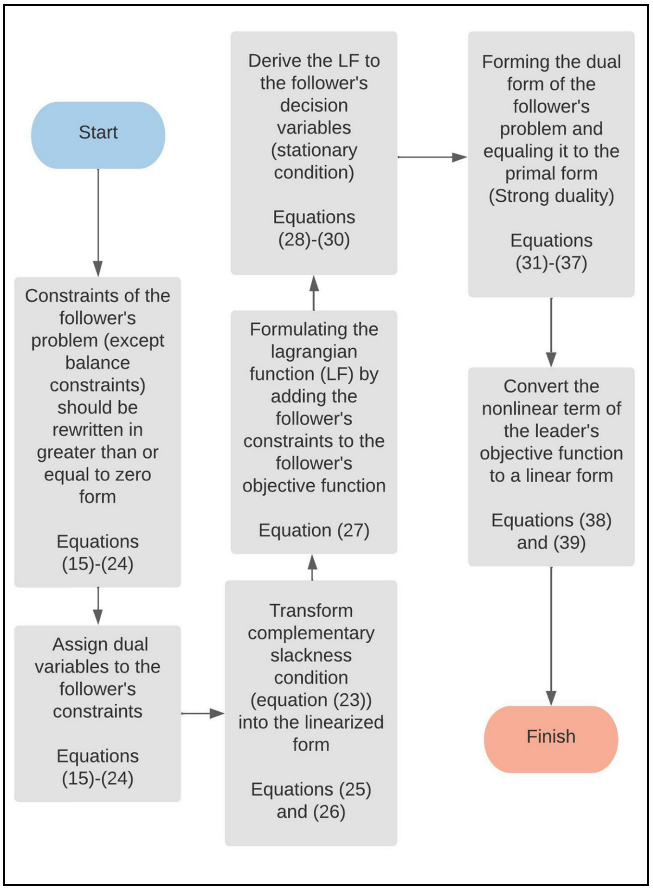


Figure 2. A brief outline of the Karush–Kuhn–Tucker method.

Multi-objective Optimization

For solving the model, we should transform the multi-objective formulation into a single mathematical programming model. For this aim, there are various popular methods in the literature, which have been used widely by different researchers. One of the well-known approaches that has proved its capability in dealing with multi-objective problems is the one introduced by Torabi and Hassini (TH method) (*26*). In this method, it is necessary to define two variables, called the positive ideal solution (PIS) and the negative ideal solution (NIS), for each objective function individually. To obtain the PIS, each objective function should be solved separately and the NIS can be achieved by the following equations:

$$Z_1^{NIS} = \text{Min}\left(Z_1\left(x_2^{PIS}\right),\ Z_1\left(x_3^{PIS}\right)\right) \tag{40}$$

$$Z_2^{NIS} = \text{Max}\left(Z_2\left(x_1^{PIS}\right),\ Z_2\left(x_3^{PIS}\right)\right) \tag{41}$$

$$Z_3^{NIS} = \text{Max}\left(Z_3\left(x_1^{PIS}\right),\ Z_3\left(x_2^{PIS}\right)\right) \tag{42}$$

It is noteworthy that the first objective function is in the form of maximization, while the second and third objective functions are of the minimization type. So, their NIS forms are formulated according to Equations 40–42. After attaining the PIS and NIS for each objective function, the membership functions can be written as follows:

$$\zeta_1(x) = \begin{cases} 1 & \text{if } Z_1 < Z_1^{NIS} \\ \frac{Z_1 - Z_1^{NIS}}{Z_1^{PIS} - Z_1^{NIS}} & \text{if } Z_1^{NIS} \leqslant Z_1 \leqslant Z_1^{PIS} \\ 0 & \text{if } Z_1 \geqslant Z_1^{PIS} \end{cases} \tag{43}$$

$$\zeta_2(x) = \begin{cases} 1 & \text{if } Z_2 < Z_2^{PIS} \\ \frac{Z_2^{NIS} - Z_2}{Z_2^{NIS} - Z_2^{PIS}} & \text{if } Z_2^{PIS} \leqslant Z_2 \leqslant Z_2^{NIS} \\ 0 & \text{if } Z_2 \geqslant Z_2^{NIS} \end{cases} \tag{44}$$

$$\zeta_3(x) = \begin{cases} 1 & \text{if } Z_3 < Z_3^{PIS} \\ \frac{Z_3^{NIS} - Z_3}{Z_3^{NIS} - Z_3^{PIS}} & \text{if } Z_3^{PIS} \leqslant Z_3 \leqslant Z_3^{NIS} \\ 0 & \text{if } Z_2 \geqslant Z_2^{NIS} \end{cases} \tag{45}$$

Finally, the tri-objective model can be converted into a single one as stated by the following formulations:

$$\max \beta(x) = \beta_0(\gamma) + (1-\gamma)\sum_h \theta_h \zeta_h(x) \tag{46}$$

$$\text{subject to } \beta_0 \leqslant \zeta_h(x)\ h = 1,2,3 \tag{47}$$

$$x \in G(x),\ \beta \text{ and } \beta_0 \in [0,1] \tag{48}$$

In the above-mentioned equations, $G(x)$ represents the feasibility region and θ_h and γ, represent the relative importance of each objective function and the balance of TH's objective function components.

Table 1. Characteristics of Numerical Examples

No.	OD size	Objective functions			Total transported by		Transported by (%)	
		Profit	Risk	Emission	Rail	Road	Rail	Road
Ex. 1	5	361,427	192,750	348,750	3450	625	85	15
Ex. 2	10	600,421	275,400	572,450	5750	2225	72	28
Ex. 3	15	701,995	227,936	606,599	6600	4955	57	43
Ex. 4	20	854,898	336,300	705,575	7900	7190	52	48
Ex. 5	25	742,477	225,525	584,775	6800	12,510	35	65

Note: OD = origin–destination.

Table 2. Characteristics and Key Parameters of the Third Numerical Example

OD name	OD number	Emission	Risk	Demand	Road price	Maximum capacity of rail	Maximum acceptable lost demand	Travel duration by	
								Rail	Road
AG	1	100	10	800	125	600	100	30	21
AI	2	80	80	1000	120	700	100	50	35
AC	3	130	10	800	120	600	100	20	14
DH	4	85	45	850	122	750	50	40	28
DR	5	110	110	1050	128	900	75	80	56
DI	6	100	20	1100	120	700	50	50	35
GL	7	90	30	800	126	950	75	50	35
GQ	8	95	95	700	122	850	100	50	35
GI	9	120	30	1100	123	800	90	20	14
LR	10	75	15	600	127	650	85	60	42
LP	11	100	20	700	120	800	100	50	35
LJ	12	80	30	800	125	800	80	30	21
QU	13	90	15	900	123	900	90	60	42
QT	14	120	80	1000	125	700	100	80	56
QJ	15	60	60	600	120	650	50	40	28

Note: OD = origin–destination.

Computational Results

In this section, for validity evaluation of the developed model and employed methods, some numerical examples are designed to demonstrate the model's behavior. After that, a real case study of the Iran railway network is chosen to be investigated thoroughly. Finally, on some critical parameters, several sensitivity analyses are carried out to indicate parameter fluctuations and their impacts on the objectives and main decisions.

The numerical experiments are performed on a personal computer with a 3.5 GHz CPU and 12.0 GB RAM, and all the programming is implemented in GAMS software. Note that all the problems are solved to optimality within a few seconds.

Numerical Example

Several hypothetical freight transportation networks as numerical examples are employed to investigate the validity and performance of the developed optimization model. The main characteristics of the numerical example are listed in Table 1. In this table, five examples with different OD sizes are represented and the related results, including objective functions, totally transported freight through both transportation modes, and percent of transportation using each mode, are depicted.

All of these examples are investigated through the network depicted in Figure 3. In Figure 3, a hypothetical freight transportation network is depicted with 23 nodes (using capital letters from A to W) and, in Table 2, several OD pairs selected from the hypothetical freight network are listed.

For further explanation, the third numerical example (Ex. 3–15 OD pairs) is selected to be investigated comprehensively. As shown in Table 3, this test problem has 15 OD pairs, in which both transportation modes (road and rail) are available and five trains are employed. Each train has a fixed cost and capacity, which are shown in Table 4, and other key parameters, including demand, road price, time, emission, and so on, are presented in Table 2.

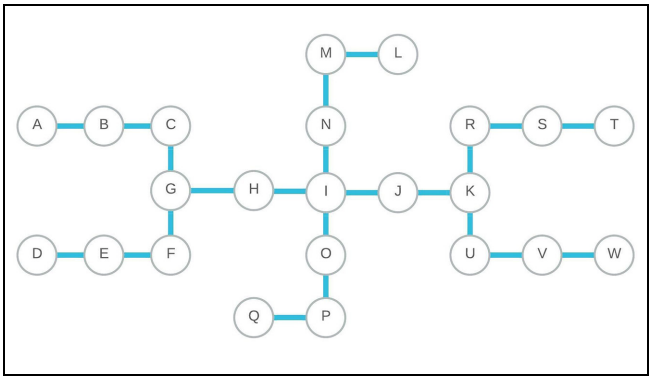


Figure 3. Schematic diagram of the hypothetical freight transportation network.

Table 3. Characteristics of Selected Numerical Example

Set name	Numbers
OD pairs	15
Trains	5
Modes of transportation	2

Note: OD = origin–destination.

Table 4. Characteristics of Trains

Train	Fixed cost for train *t*	Capacity of train *t*
1	11	600
2	15	750
3	21	950
4	15	650
5	20	900

In Table 5, the price by railway operator is proposed, considering the road price. In addition to the price, Table 5 specifies other key characteristics, including the freight transported by each transportation mode and lost demand. Figure 3 shows the differences between the proposed price by the rail operator and road systems. As is obvious from Table 5 and Figure 4, despite the lower price of rail transportation, in some OD pairs, the shipper uses road transportation because of time priority or the capacity limitations of rail transportation. Table 6 demonstrates the number of required trains of each type used for freight transportation.

As is obvious from Figure 4, even any small changes in rail prices can decrease the leader's total profit. In fact, an equilibrium is achieved through the bi-level approach in which any deviation from this optimal point (whether an increase or decrease in rail price) can result in a decrease in profit. According to Figure 5, increasing

Table 5. Results Extracted From Solving Selected Numerical Example

OD name	OD number	Road price	Rail price	Freight transported by	
				Rail	Road
AG	1	125	116	600	100
AI	2	120	105	600	300
AC	3	120	114	600	100
DH	4	122	110	750	50
DR	5	128	104	800	175
DI	6	120	105	700	350
GL	7	126	111	725	0
GQ	8	122	107	375	225
GI	9	123	117	800	210
LR	10	127	109	515	0
LP	11	120	105	600	0
LJ	12	125	116	720	0
QU	13	123	105	810	0
QT	14	125	101	440	460
QJ	15	120	108	550	0

Total transported freight by	Quantity
Rail	6600
Road	4955
Cost	701,995
Risk	227,936
Emission	606,599

Note: OD = origin–destination.

the optimal price proposed by the leader has more influence than decreasing the rail price on the leader's total profit. The rail operator should make a conscious decision about pricing policy, as any inappropriate decision can lead to a loss in market share.

In Figure 5, the pair numbers in parentheses show the percentage of freight transported by rail and road transportation modes, respectively. As is obvious, at the optimal point, the rail system has gained the maximum ratio of freight transportation. Increasing the optimal price of rail transportation not only reduces the total profit but also decreases the ratio of freight transported by the rail system and, on the other hand, increases the ratio of transported freight associated with the road system. The green and red numbers in parentheses represent the rail and road ratio of freight transportation, respectively.

Case Description

As mentioned earlier, the railway operator owns trains of various types and capacities, operating on the paths shown in the following map. Figure 6 demonstrates the main structure of the railway system in Iran in which different OD pairs act as different demands. As is obvious, 28 stations are designated by different cities' names to be operational for 100 OD pairs. It is noteworthy that these

Table 6. Required Trains of Each Type

Train type	Numbers of required trains
1	3
2	3
3	2
4	3
5	2

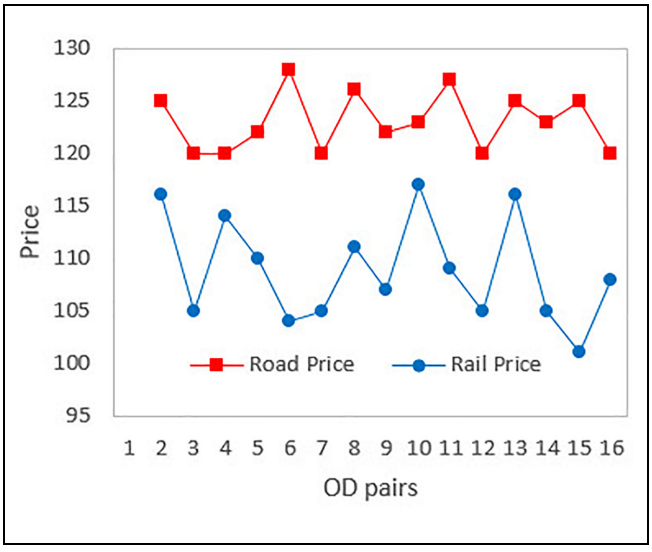


Figure 4. Differences between the road and rail prices.
Note: OD: origin–destination.

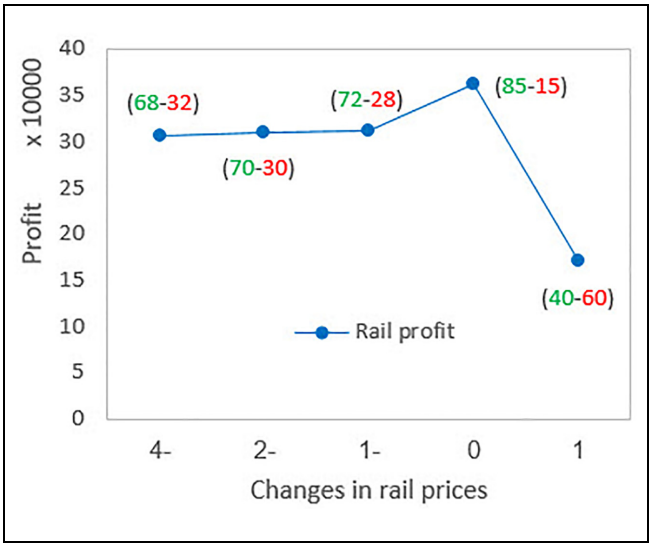


Figure 5. Effect of rail prices on total profit. (Color online only.)

demands (OD pairs) are satisfied through nine various trains and the obtained results reveal that the current rail share for freight transportation is 30% of total freight

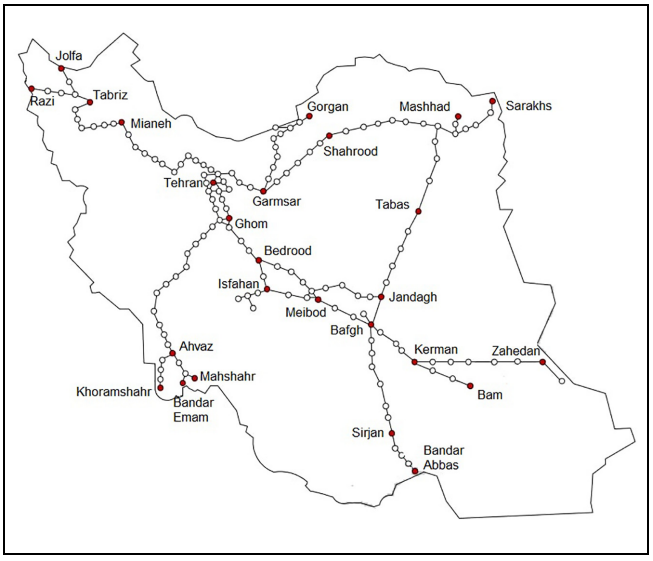


Figure 6. Iran railway network (27).

transportation. In fact, the shipper desires to satisfy the demands from origins to destinations in an efficient and timely manner.

For example, an OD demand like Jolfa–Tehran can be met by rail or road transportation modes or a combination of those. In the following section, sensitivity analyses are done on the parameters obtained from this case study. In addition to analyses of these parameters, some managerial insights for improvement are provided.

Sensitivity Analysis

In this section, we investigate the impact of some major parameters on the various leader's objective functions. The effects that variation in these parameters produce are shown in the following tables.

Line capacity variation is examined in Table 7. As is obvious from Table 7, the greater the line capacity, the greater the market share the rail operator could absorb.

Increasing the line capacity by decision-makers in the rail system can enhance rail profitability. To this end, investment in railway infrastructure can lead to more profit in the long run and railway system authorities should take into account this decisive factor. The rail operator can invest to purchase new fleets, add them to the transportation network, and increase the frequency of trains between different origins and destinations, consequently making the rail system a dominant mode of freight transportation. Therefore, when a situation arises in which the shipper an select between rail and road systems for meeting their demand, choosing the rail system can be a more probable option. Other attempts that can be made for improvement in line capacity by rail

Table 7. The Effect of Line Capacity Variation

Line capacity	OD pair	Objective functions			Quantity transported (%)	
		Profit	Risk	Emission	Rail	Road
−50%	100	1,421,897	448,500	946,000	16	84
−25%	100	2,094,989	706,121	1,347,394	24	76
0	100	2,649,067	905,550	1,692,100	30	70
+ 25%	100	3,276,271	1,170,281	2,168,481	38	62
+ 50%	100	3,794,622	1,406,350	2,560,100	44	56

Note: OD = origin–destination.

Table 8. The Effect of Train Capacity Variation

Train capacity	Objective functions			Transported freight		%	
	Profit	Risk	Emission	Rail	Road	Rail	Road
−50%	1,430,897	462,000	978,000	13,350	69,415	16	84
−25%	2,051,111	676,058	1,318,808	19,168	63,597	23	77
0	2,649,067	905,550	1,692,100	24,850	57,915	30	70
+ 25%	3,230,862	1,150,650	2,102,890	30,567	52,198	37	63
+ 50%	3,774,761	1,384,200	2,501,950	36,075	46,740	44	56

decision-makers are the rehabilitation of worn-out railways and the construction of double-track railways.

In addition to line capacity, according to Table 8, the impact of train capacity expansion is shown. Employing trains with higher capacity not only can absorb freight from the shipper but also can augment the profitability of the railway system. Results obtained from Tables 7 and 8 reveal that key infrastructure such as line and train capacity are of great importance for increasing the profitability of the rail system, and therefore full attention should be paid to this issue by rail operators. Obsolete wagons should be replaced with more modern and capacious ones to transport more freight volume on each trip. In this regard, train make-up with more capacious wagons and up-to-date freight loading and unloading equipment can be taken into consideration to persuade the shipper to select the rail option.

The shipper selects their transportation modes based on time and price factors. For this aim, in this paper parameter α can be used as a tuning parameter between time and price priority for the shipper. As depicted in Figure 7, an increase in α means an increase in the tendency to prioritize the price factor by the shipper.

In other words, the greater the value of α, the more willing the shipper is to deliver in a low-cost manner. In fact, increasing the value of α leads to the tendency of the shipper to employ rail transportation modes and this also gives rise to an escalation trend in all objective functions. This means that increases in the value of α not only increase the rail sector's total profit but also lead to more risk and emissions in transportation. Table 9 shows the details of increasing parameter α and its effect on objective functions and freight transported by each transportation mode.

Figure 8 demonstrates the impact of the unit penalty cost on the total lost demand. In this figure, as can be observed, increasing the unit penalty cost exerts intense pressure on the shipper to transport their freight through

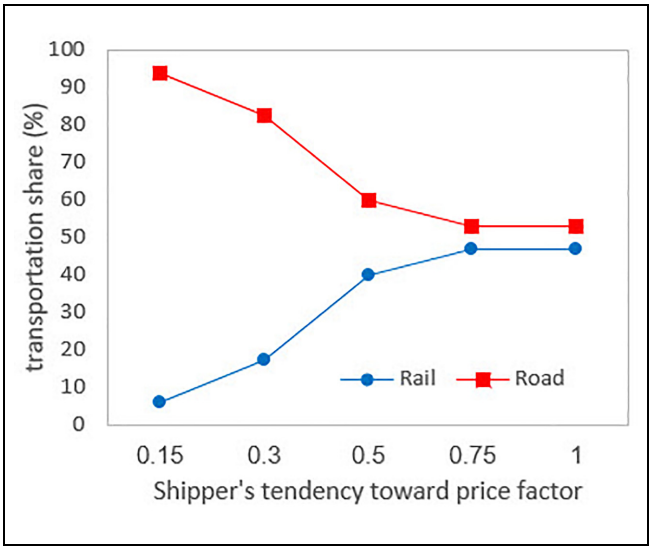


Figure 7. Trade-off between price and time for shippers.

Table 9. The Effect of Shipper's Tendency Toward Price and Time

	Objective functions			%	
α	Profit	Risk	Emission	Rail	Road
0.15	1,448,145	846,925	1,751,000	11	89
0.3	2,248,239	869,975	1,753,575	21.5	78.5
0.5	2,649,067	905,550	1,692,100	42.5	57.5
0.75	2,901,177	882,600	1,749,450	48	52
1	3,039,123	870,625	1,692,850	48	52

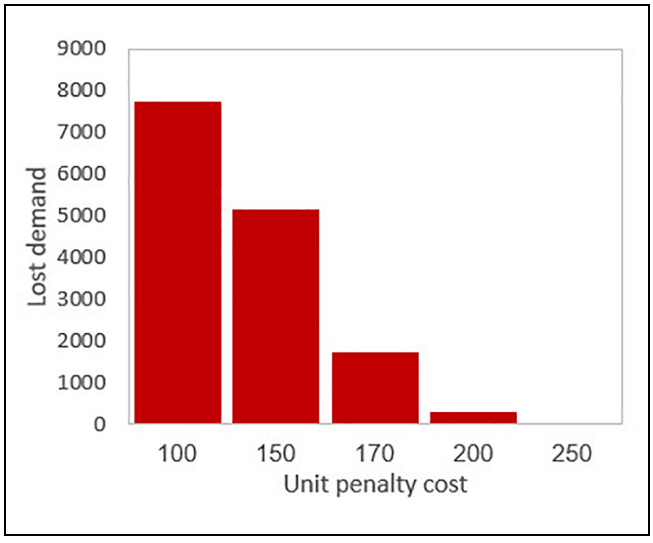


Figure 8. Penalty cost effect on lost demand.

transportation modes and make an effort to deliver the OD pair related freight demands totally.

Table 10 shows the impact of the penalty cost for unsatisfied demand on objective functions and lost demand thoroughly. A higher penalty cost forces the shipper to decrease unsatisfied demand significantly and it is evident that the higher the penalty cost, the lower the unmet demand will be. As is obvious from the last two columns in Table 10, increasing penalty cost for the lost demand persuades the shipper to shift from rail to road, which means that the shipper is forced to meet the demands as much as possible. For the shipper, choosing the road sector for satisfying demand can be a better option compared to the rail one. In other words, trying to achieve the least level of lost demand stimulates the shipper to choose the road sector to reach a high level of customer satisfaction.

Two decisive parameters influencing the results derived from the multi-objective method used in this paper are θ_h and γ. The impacts of these two pivotal parameters are depicted in Table 11 and Figure 9. When γ holds a lower value, the weights of the objective functions play key roles in specifying the objective values. As demonstrated in Table 11, when γ increases in value, the second and third objective functions (minimization in risk and emission) show better performance and a downward trend, while the first objective worsens (maximization in profit). In this paper, maximization in profit is of great importance for the rail decision-maker, so its weight is considered higher in value compared to the second and third objective functions. With increasing γ, the tendency to profit maximization is decreased by decision-makers, and freight transported by the road system finds an increasing trend.

As described earlier, Figure 9 is related to the relative weights of the importance of the objective functions. In this figure, the horizontal axes describe the relative weight of the second and third objective functions (risk and emission) in comparison with the first objective function (profit). It is obvious, in Figure 9*a*, that by increasing the second objective weight, the first objective function is exacerbated while the second objective function holds better values. Similar to Figure 9*a*, in Figure 9*b*, the first and third objective functions are compared based on their relative importance weights. Increasing the third objective function's weight leads to lower values for the both the first and third objective functions. In fact, these two figures show situations in which the preferences of decision-makers are changed and various values for objective functions are obtained. Adoption of an eco-friendly approach, that is, a tendency toward reduction of carbon emissions by the carrier, leads to a marked drop in total profit, which can be a contributing factor to force the carrier to employ cleaner trains that are more environmentally friendly compared to obsolete ones. This gradual transition from outdated equipment to green equipment not only gives rise to fewer environmental issues but also can absorb more customers and consequently more profit in the long run.

Conclusion and Future Directions

Through this study, we aimed to investigate a freight transportation system including rail and road transportation modes by employing a mathematical bi-level

Table 10. The Effect of Unit Lost Penalty Cost on Lost Demand

		Objective functions			Transported freight		%	
Lost penalty	Lost	Profit	Risk	Emission	Rail	Road	Rail	Road
100	7725	2,643,701	910,775	1,716,425	24,800	51,925	32	68
150	5120	2,650,231	901,575	1,719,175	24,800	54,530	31	69
170	1685	2,649,067	905,550	1,692,100	24,850	57,915	30	70
200	265	2,652,161	891,750	1,704,000	24,800	59,385	29	71
250	0	2,652,161	891,750	1,704,000	24,800	59,650	29	71

Table 11. The Impact of Multi-objective Method Parameters

	Objective functions			Transported freight		%	
γ	Profit	Risk	Emission	Rail	Road	Rail	Road
0.1	2,649,067	905,550	1,692,100	24,850	57,915	30	70
0.2	2,594,478	835,752	1,549,972	24,500	58,265	30	70
0.3	2,514,564	766,950	1,422,372	23,850	58,915	29	71
0.4	2,416,832	713,098	1,322,501	23,300	59,465	28	72
0.5	1,654,080	432,078	801,325	16,165	66,600	20	80
0.6	1,653,421	432,348	801,827	16,158	66,656	20	80
0.7	1,654,080	432,078	801,325	16,165	66,600	20	80

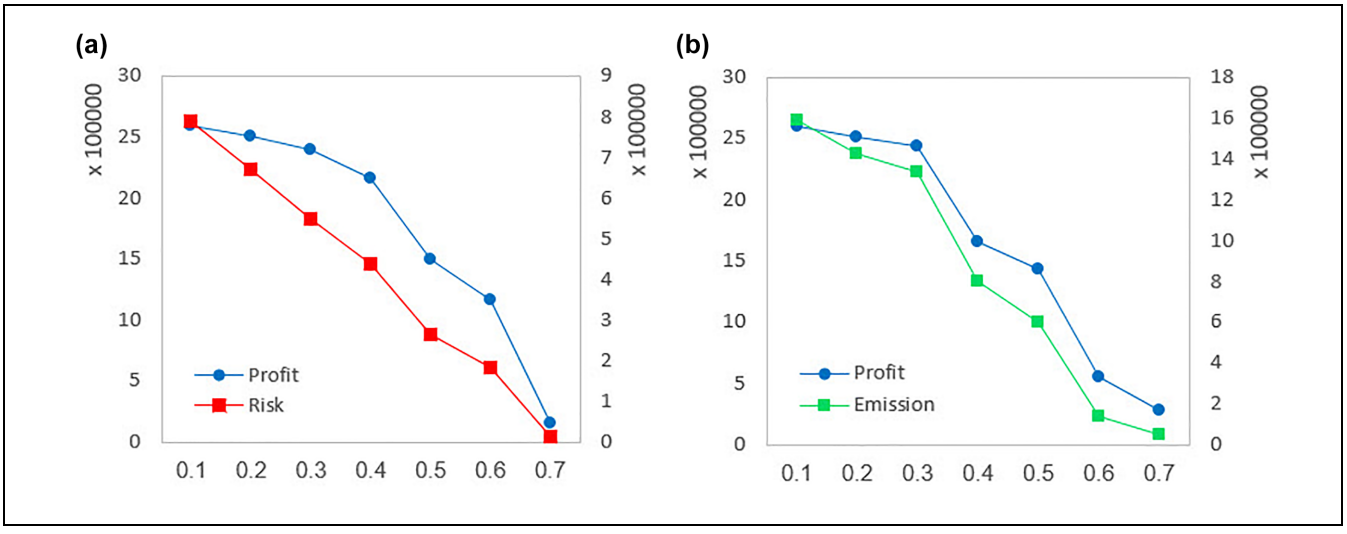


Figure 9. Interaction between objective functions: (*a*) interaction between profit and risk and (*b*) interaction between profit and emission.

programming model. In an attempt, the triple bottom line (TBL) in sustainable transportation context, namely, economic, environmental, and social aspects, are explored simultaneously in the developed model. Furthermore, a bi-level programming model is used to simulate the interactions between the rail system and the shipper. The shipper's tendency toward time or cost aspects can influence mode choice or modal shift significantly. Several sensitivity analyses are carried out on key parameters, and their effects on objective functions and freight transported by the rail or road systems are evaluated.

Future directions can be considered as an extension of game theory methods, such as Bertrand or Cornout games, considering a game at the upper level between two leaders including rail and road systems and shippers in the lower level. Using uncertainty programming for dealing with uncertainty in some critical parameters,

such as demand, travel time, costs, and so on, can be highly beneficial. Another extension can be related to taking into account the concept of quantity discount, which can be considered for different modes.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: Sh. Rahiminia, A. Mehrabi; data collection: Sh. Rahiminia, A. Mehrabi; analysis and interpretation of results: Sh. Rahiminia, A. Mehrabi, M.P. Aghaee; draft manuscript preparation: Sh. Rahiminia, A. Mehrabi, M.P. Aghaee, A. Jamili. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

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